

ABC tells us *how important* an SKU is (A = high value → control tightly).

XYZ tells us *how stable* the demand is (X = stable → easy to forecast; Z = erratic → hard to forecast).

Together they create the famous **ABC-XYZ matrix** (9 categories). The golden insight: **AX ≠ CZ** — you treat them completely differently in inventory, procurement, and planning.

1. Learn: Why Demand Variability Matters + Coefficient of Variation (CV)

Demand variability = how much the actual demand fluctuates period-to-period (month, week, etc.).

- **Stable demand** → you can forecast accurately → low safety stock → lower inventory cost.
- **Erratic demand** → forecasts are unreliable → you need higher safety stock or different strategies (or even stop stocking the item).

The standard statistical measure used in XYZ analysis is the **Coefficient of Variation (CV)**:

$$CV = \frac{\sigma}{\mu}$$

where:

- σ \sigma σ = standard deviation of demand (measures spread)
- μ \mu μ = mean (average) demand over the historical periods

Interpretation:

- CV close to 0 → very stable (demand almost the same every period)
- CV = 0.5 → std dev is 50 % of the mean → noticeable fluctuation
- CV > 1 → std dev bigger than the mean → extremely erratic

Important notes (from real-world practice):

- Use at least 12 periods (months) of clean historical demand data.
- Use sample standard deviation (ddof=1 in Python, STDEV.S in Excel).
- Thresholds are **not universal** — they are business-specific. We will use the common practical set:
 - **X** = CV ≤ 0.25 → stable / predictable
 - **Y** = 0.25 < CV ≤ 0.5 → moderate variability
 - **Z** = CV > 0.5 → erratic / highly variable (Some companies use 0.1 / 0.25 or 0.5 / 1.0 — you can adjust later based on your data distribution.)

5. Insight: AX ≠ CZ (Treat them VERY differently)

Category	Value	Predictability	Recommended Strategy (real-world)
AX	High	Very high	JIT / low safety stock, frequent small orders, tight MRP control, high service level target (99 %)
AY	High	Moderate	Moderate buffers, periodic review
AZ	High	Very low	High safety stock or redesign product / find alternative supplier / make-to-order
BX	Medium	High	Standard replenishment
BY	Medium	Moderate	Slightly higher buffers
BZ	Medium	Low	Review ordering frequency
CX	Low	High	Bulk orders, minimal monitoring
CY	Low	Moderate	Minimal stock
CZ	Low	Very low	Highest risk — consider eliminating the SKU, vendor-managed inventory, or very high buffers only if critical

Key takeaway:

- AX items → you can run very lean (saves cash).
- CZ items → they eat up working capital and still cause stock-outs → many companies actively reduce or eliminate CZ SKUs.